

2_SQL Indexing Exercises

Exercise 1: Non-Clustered Index

Answer:

```
CREATE NONCLUSTERED INDEX IX_Products_ProductName ON Products(ProductName);
```

Exercise 1 Output

```
SELECT * FROM Products WHERE ProductName='Laptop';
```

```
ProductID:1 ProductName:Laptop  
Category:Electronics Price:1200.00
```

```
Before Index: 15 ms
```

```
After Index : 2 ms
```

Exercise 2: Clustered Index

Answer:

```
CREATE CLUSTERED INDEX IX_Orders_OrderDate ON Orders(OrderDate);
```

Exercise 2 Output

```
SELECT * FROM Orders WHERE OrderDate='2023-01-15';
```

```
OrderID:1 CustomerID:1  
OrderDate:2023-01-15
```

```
Before Index: 12 ms
```

```
After Index : 1 ms
```

Exercise 3: Composite Index

Answer:

```
CREATE INDEX IX_Orders_CustomerID_OrderDate ON Orders(CustomerID, OrderDate);
```

Exercise 3 Output

```
SELECT * FROM Orders WHERE CustomerID=1 AND OrderDate='2023-01-15'
```

```
OrderID:1 CustomerID:1  
OrderDate:2023-01-15
```

```
Before Index: 10 ms
```

```
After Index : 1 ms
```